Geometric Contruction of Brillouin Zones as Voronoi Cells on Reciprocal Lattices for Phonon Propagation

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1 Introduction

The Brillouin zone is an important concept in crystallography and solid-state physics, representing the fundamental region in reciprocal space (k-space or momentum space) that contains all the unique modes of wave propagation in a crystal lattice. The wavevector for a phonon k in k-space determines how a phonon in real space moves throughout a given lattice. The Brillouin Zone can be constructed geometrically as the Voronoi cell of origin in the reciprocal lattice. This project aims to explore the geometric construction of Brillouin zones and understand dynamic phonon propagation through a geometric interpretation of the wavevector k in k-space.

2 Background

2.1 Voronoi Diagrams and Delaunay Triangulations

Voronoi diagrams and Delaunay triangulations are fundamental concepts in computational geometry that are closely related to each other [2, 4]. A Voronoi diagram is a partioning of a plane into regions based on the distance to a specific set of points. Specifically, any point within a given region is closer to its corresponding point than to any other point in the set. A voronoi cell for a given point p_i, denoted $V(p_i)$, can be defined as follows:

$$V(p_i) = \{q \in \mathbb{R}^n | ||q - p_i|| \leq ||q - p_j|| \text{ for all } j \neq i\} \tag{1}$$

An example of a Voronoi diagram in $\mathbb{R}^2$ is shown in Figure 1, where the points p_i are shown as black dots, and the Voronoi cells are shown as colored regions.

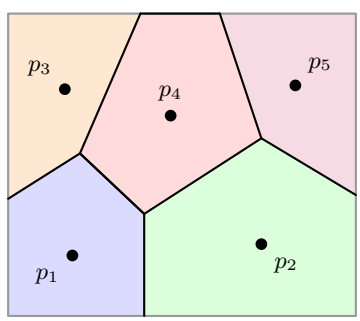

Figure 1: A Voronoi diagram in $\mathbb{R}^2$ for five sample sites. Each shaded cell contains the points closest to its generating site.

It will be important to note that Voronoi diagrams can be defined in any number of dimensions, however, we will be specifically interested in Voronoi diagrams in $\mathbb{R}^3$ for the purpose of this project.

There are several algorithms for computing these diagrams directly, including Fortune's algorithm and its beach-line sweep, however this type of algorithm does not easily generalize to higher dimensions. One option for computing Voronoi diagrams in higher dimensions is to use a divide and conquer approach (CITE Divide and conquer paper), however, the most common approach is to compute the Delaunay triangulation of the given point set and then extract the Voronoi diagram from the Delaunay triangulation [1]. The Delaunay triangulation is a triangulation of a set of points that satisfies the Delaunay condition, which enforces that no point in the set can exist inside the circumcircle/sphere of any triangle/tetrahedron in the triangulation. A convenient property of Delaunay triangulations is that they are the dual of Voronoi diagrams, a property that is generalizable to any number of dimensions [1]. An example of a Delaunay triangulation in $\mathbb{R}^2$ is shown in Figure 2, where the points p_i are shown as black dots, and the Delaunay triangulation is shown as a network

of edges connecting the points. Note that this Delaunay triangulation is the dual of the Voronoi diagram shown in Figure 1.

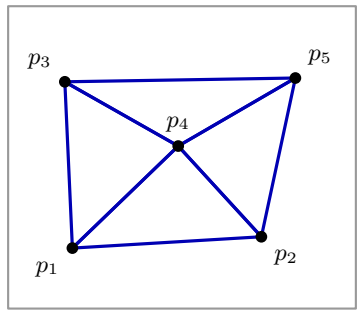

Figure 2: The Delaunay triangulation dual to Figure 1. Two sites are connected when their Voronoi cells share an edge.

Since extracting the dual from a triangualtion is straightforward relative to the construction of the triangulation itself, we can compare the complexity of computing the Voronoi diagram to the complexity of computing the Delaunay triangulation, in which they are similar.

2.2 Phonons

Phonons are quasi-particles that represent quantized vibrational modes in a crystal lattice. This is to say that they are a way to describe the collective vibrations of atoms in a solid as if they were particles. You can think of phonons as quantum mechanical versions of classical sound waves, and as sound waves can be visualized as waves in a given medium, phonons can be visualized as wave/particles traveling through a crystal lattice. It is crucial to be able to study phonons in order to understand various properties of materials and is very important in the field of condensed matter physics.

2.3 Reciprocal Lattices and Brillouin Zones

A crystal lattice is a periodic arrangement of atoms in $\mathbb{R}^3$, which can be described by a set of basis vectors. A given lattice L can be represented then as:

$$L = \{n_1 a_1 + n_2 a_2 + n_3 a_3 | n_i \in \mathbb{Z}\} \tag{2}$$

where a_i are the basis vectors of the lattice.

A reciprocal lattice is a mathematical construct used in the study of periodic structures, crystal lattices being prime examples. It is derrived from the real-space lattice and is used to analyze wave phenomena in crystals. To get the reciprocal lattice, you can take the Fourier transform of the real-space lattice, which alludes to why this lattice lives in momentum space. It is important to note that the reciprocal lattice for a periodic structure is also periodic, exhibiting translational symmetry (i.e. shifting the lattice by by a reciprocal lattice vector leaves the lattice unchanged). We can represent the reciprocal lattice L^* as:

$$L^* = \{m_1 b_1 + m_2 b_2 + m_3 b_3 | m_i \in \mathbb{Z}\} \tag{3}$$

where b_i are the basis vectors of the reciprocal lattice, which can be calculated from the basis vectors of the real-space lattice, where:

$$b_i \cdot a_j = 2\pi \delta_{ij} \tag{4}$$

which means that the basis vectors of the reciprocal lattice are orthogonal to the basis vectors of the real-space lattice unless $i = j$, in which case they are parallel and have a magnitude of 2π.

The First Brillouin zone (often just called the Brillouin Zone) is a fundamental region of k-space defined as the Wigner-Seitz cell of the reciprocal lattice, which gemetrically is represented as the Voronoi cell of the origin in the reciprocal lattice (where the lattice vector G is zero). Due to the aforementioned periodicity (and subsequent translational symmetry) of the reciprocal lattice, the Brillouin zone contains all the unique wavevectors that can describe phonon propagation in the crystal lattice (all allowed values of k). The Brillouin zone is important for understanding the behavior of phonons in a crystal lattice, such as their dispersion relations and subsequently their group velocities, which are crucial for understanding thermal and electrical properties of materials.

3 Computing Voronoi Diagrams in $\mathbb{R}^3$

There are two algorithms we will discuss for computing Voronoi diagrams, which reduces to the method of computing the Delaunay triangulation in $\mathbb{R}^3$ of the given point set $P \in \mathbb{R}^3$ (the reciprocal lattice points). The first of which is the incremental construction algorithm for Delaunay Triangulations, a more intuitive approach, and the second of which is the lifting approach, which involves lifting the points in $\mathbb{R}^3$ to $\mathbb{R}^4$ and then computing the convex hull of the lifted points. For the three-dimensional setting and the extraction of the dual structure, our discussion follows the perspective described by Ledoux [1].

3.1 Predicates

3.1.1 Orientation Test

The orientation test is a fundamental predicate used in widely in computational gemoetry algorithms, including those for computing Delaunay triangulations [2, 4]. The purpose of an orientation test is to determine the relative orientation of a set of points in space. In $\mathbb{R}^3$, the orientation test for four points a, b, c, d can be defined as follows:

$$\textsc{Orient}(a, b, c, d) = \text{sign} \begin{vmatrix} a_x & a_y & a_z & 1 \\ b_x & b_y & b_z & 1 \\ c_x & c_y & c_z & 1 \\ d_x & d_y & d_z & 1 \end{vmatrix} \tag{5}$$

In short, the orientation test computes the determinant of a matrix formed by the coordinates of the four points, and the determinant's sign indicates the relative orientation of the points. Our primary use of this orientation test will be to determine whether a point lies above, below, or on a plane defined by the three other points.

3.1.2 In-Sphere Test

The in-sphere test is another fundamental predicate used in computational geometry algorithms, and is constructed in a similar way to the orientation test. The in-sphere test for five points a, b, c, d, e in $\mathbb{R}^3$ can be defined as follows:

$$\textsc{InSphere}(a, b, c, d, e) = \text{sign} \begin{vmatrix} a_x & a_y & a_z & a_x^2 + a_y^2 + a_z^2 & 1 \\ b_x & b_y & b_z & b_x^2 + b_y^2 + b_z^2 & 1 \\ c_x & c_y & c_z & c_x^2 + c_y^2 + c_z^2 & 1 \\ d_x & d_y & d_z & d_x^2 + d_y^2 + d_z^2 & 1 \\ e_x & e_y & e_z & e_x^2 + e_y^2 + e_z^2 & 1 \end{vmatrix} \tag{6}$$

The reason why this test contains 5 points is because a sphere in $\mathbb{R}^3$ is defined by 4 points, and the in-sphere test determines whether the fifth point e lies inside, outside, or on the sphere defined by the first four points a, b, c, d. The in-sphere test is crucial for maintaining the Delaunay condition during the construction of the Delaunay triangulation.

3.2 Half-Faces and their Alignment

In order to reliably use our defined predicates (orientation test/in sphere test), we must have a consistent ordering for vertices in faces. The order of vertices determines the result of the two tests (i.e. the way we construct the matrix) and thus we have to decide on a consistent convention. There are two ways that we

will ensure this. The first, is to utilize the concept of "half-faces", which is the 3 dimentional version of a "half-edge". When we store the triangulation, we store each tetrahedron. Any tetrahedron can be constructed with 4 vertices, which we use to generate each half-face of the tetrahedron. Each half-face of the tetrahedron points outward, which can be ensured by performing an orientation test on the face to be constructed and the last remaining point in the Tetrahedron, and flipping vertices in the ordering until it returns < 0. Each half-face can be the pair to another half-face, which represents two tetrahedra being adjacent along that face. In this case, both half-faces store a pointer to its pair. This enables both the efficient traversal between tetrahedra, and also the quick reconstructing of new tetrahedra during flipping, as faces can be reused from old tetrahedra that already point to relevant adjacent tetrahedra through its adjacent half-face.

3.3 Incremental Construction Algorithm for Delaunay Triangulations

The incremental construction algorithm for Delaunay triangulations involves the insertion of points one at a time into an existing triangulation, and then updating the triangulation to maintain the global Delaunay property. Maintaining the Delaunay property involves a process called edge flipping, which is a local operation that replaces one edge of a triangulation with another edge to restore the Delaunay property. For the purpose of analysis, we will use randomized incremental construction, which involves inserting the points in a random order.

To describe the algorithm, we will first define a few different functions that we will use in the algorithm. The first is a function that identifies the tetrahedron in the current triangulation that contains a given point p. The second is a function that checks whether a given tetrahedron violates the Delaunay condition, and the third is a function that

To begin, we define the function FindContainingTetrahedron (Algorithm 1) that takes in a Delaunay triangulation and a point, locating the tetrahedron in the triangulation that contains the point. This is done by starting with an arbitrary tetrahedron and then traversing the triangulation until we find the correct tetrahedron. This style of walk-based point location is closely related to the approach analyzed by Mücke, Saias, and Zhu [3].

Then we define the function UpdateDelaunay (Algorithm 2) that takes in a Delaunay triangulation and a new point to be inserted, and updates the triangulation to include the new point while maintaining the Delaunay property. This utilizes our previously defined function FindContainingTetrahedron to find the tetrahedron that contains the new point, and then performs edge flips as necessary to restore the Delaunay property.

Finally, we can describe the overall Delaunay construction procedure itself. This routine begins with a super tetrahedron containing all input points, inserts the points one by one using Algorithm 2, and then removes any tetrahedra incident to the vertices of the super tetrahedron.

3.4 Computing Voronoi Diagrams from Delaunay Triangulations

As mentioned previously, the Voronoi diagram of a set of points can be computed from the Delaunay triangulation of those points. This is because the dual of a Delaunay triangulation is the Voronoi diagram on the same set of points P. We want to extract 4 important pieces of information from the Delaunay triangulation in order to construct the Voronoi diagram: the vertices, edges, faces, and polyhedra (the cells) of the Voronoi diagram. We can extract the vertices of the Voronoi diagram from the circumcenters of the tetrahedra in the Delaunay triangulation, the edges of the Voronoi diagram from the shared faces of adjacent tetrahedra, the faces of the Voronoi diagram from the edges of the Delaunay triangulation, and the cells of the Voronoi diagram from the vertices of the Delaunay triangulation [1].

3.5 Data Structures for Representing Delaunay Triangulations and Voronoi Diagrams

It is important to note that the construction of a Voronoi diagram is not necessarily required in full. Rather, we can explicitly construct the Delaunay triangulation and simply query the necessary information from the triangulation using operations similar to those described in Algorithm 4 for targeted queries. For example, if we only want a single polygon in the Voronoi diagram, we can simply query the Delaunay triangulation for

the necessary information to construct that single polygon without having to construct the entire Voronoi diagram. This idea will motivate why we will utilize a data structure most relevant for representing the Delaunay triangulation, as opposed to the Voronoi diagram. We will however store entire Voronoi diagrams for visualization purposes, but these need not be fast to construct, as they will not be used for querying.

Since the triangulation is in $\mathbb{R}^3$, we cannot use a DCEL (Doubly Connected Edge List) to represent the triangulation, as it is only suitable for representing planar subdivisions. Instead, we will store all tetrahedra in the triangulation, and for each tetrahedron, we will store pointers to its four vertices, its four faces, and its four adjacent tetrahedra. This way, we can easily navigate the triangulation and extract the necessary information for computing the First Brillouin zone as the Voronoi cell of the origin in the reciprocal lattice.

3.6 Taking the Dual of a Delaunay Triangulation

To take the dual of a Delaunay triangulation, we take advantage of the method described by [1], which describes how to extract the vertices, edges, faces, and cells of the Voronoi diagram from the Delaunay triangulation.

Let T be the DT of a set S of points in R3. The simplices of the dual D of T can be computed as follows (all the examples refer to Figure 3:

Vertex: a single Voronoi vertex is easily extracted:it is located at the centre of the sphere passing through the four vertices of its dual tetrahedron τ.

Edge: a Voronoi edge, which is dual to a triangular face κ, is formed by the two Voronoi vertices dual to the two tetrahedra sharing κ.

Face: a Voronoi face, which is dual to a Delaunay edge, is formed by all the vertices that are dual to the Delaunay tetrahedra incident to α. The idea is simply to 'turn' around a Delaunay edge and extract all the Voronoi vertices. These are guaranteed to be coplanar, and the face is guaranteed to be convex.

Polyhedron: the construction of one Voronoi cell Vp dual to a vertex p is similar: it is formed by all the Voronoi vertices dual to the tetrahedra incident to p. Since a Voronoi cell is convex by definition, it is possible to collect all the Voronoi vertices and then compute the convex hull (e.g. see [3] for an algorithm); the retrieval of all the tetrahedra incident to p can be done by performing a BFS-like algorithm on the graph dual to the tetrahedra. A simpler method consists of first identifying all the edges incident to p, and then extracting the dual face of each edge.

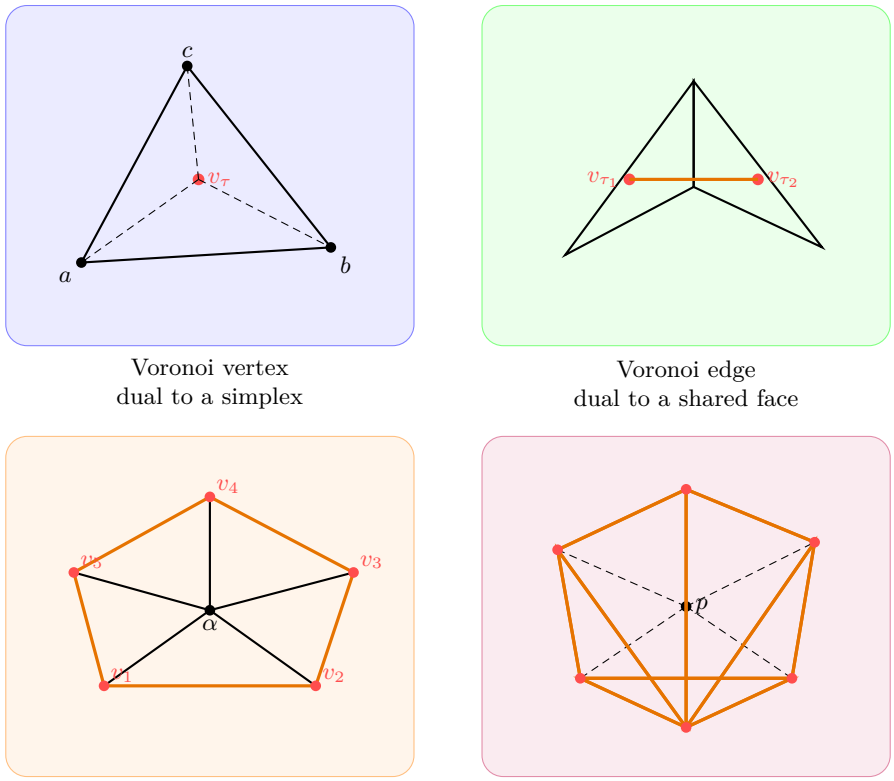

Figure 3: Duality between Delaunay simplices and Voronoi simplices. A Delaunay simplex gives a Voronoi vertex, a shared Delaunay face gives a Voronoi edge, a Delaunay edge gives a Voronoi face, and a Delaunay vertex gives an entire Voronoi cell.

3.7 Setting up the Reciprocal Lattice and Computing the Brillouin Zone

References

[1] Hugo Ledoux. Computing the 3D Voronoi Diagram Robustly: An Easy Explanation. In *4th International Symposium on Voronoi Diagrams in Science and Engineering (ISVD 2007)*, pages 117–129.

[2] David Mount. Lecture Notes - David Mount.

[3] Ernst P Mücke, Isaac Saias, and Binhai Zhu. Fast randomized point location without preprocessing in two- and three-dimensional Delaunay triangulations.

[4] Ka Yaw Teo. Lecture Notes - Ka Yaw Teo.

A Algorithms

Algorithm 1 Finding the Tetrahedron Containing a Point in a Delaunay Triangulation

Input: A Delaunay triangulation T of a point set P in $\mathbb{R}^3$, and a point p to be located.

Output: The tetrahedron t in T that contains the point p.

```
procedure FindContainingTetrahedron(T, p)
    let t be an arbitrary tetrahedron in T.
    let V be the set of visited Tetrahedra
    let out = None
    while true do
        if t ∈ V then
            Return None
        end if
        Add t to V
        for each face f_i of t do
            if  thenOrient(f_i[0], f_i[1], f_i[2], p) > ε
                Set out to f_i
                break
            end if
        end for
        if out = None then
            Return t
        else
            if the pair of out does not exist then
                Return None
            else
                Let t be the tetrahedron of the pair of out
                Set out = None
            end if
        end if
    end while
end procedure
```

Algorithm 2 Insertion of a Point into a Delaunay Triangulation

Input: A current Delaunay triangulation T of a point set P in $\mathbb{R}^3$, and a new point p to be inserted.

Output: An updated Delaunay triangulation T' that includes the new point p.

```
procedure UpdateDelaunay(T, p)
    Find the tetrahedron t in T that contains p.
    Subdivide t into four new tetrahedra by connecting p to the vertices of t.
    Reconnect the new tetrahedra to the adjacent tetrahedra of t.
    Connect pairs of new tetrahedra that share a face.
    Remove t from T and add the four new tetrahedra to T.
    Initialize a queue Q with the new tetrahedra.
    while Q is not empty do
        Dequeue a tetrahedron t' from Q.
        if t' is no longer in T then
            continue
        end if
        for each face f of t' do
            if the pair of f exists then
                let t'' be the tetrahedron of the pair of f
                let v be the vertex of t'' opposite to f
                if v lies strictly inside the circumsphere of t' then
                    Perform a local flip between t' and t''.
                    Enqueue the new tetrahedra formed by the flip into Q.
                    break
                end if
            end if
        end for
    end while
    return the updated triangulation T'.
end procedure
```

Algorithm 3 Constructing a Delaunay Triangulation

Input: A point set P in $\mathbb{R}^3$.

Output: A Delaunay triangulation T of P.

```
procedure ConstructDelaunay(P)
    Construct a super tetrahedron T_large containing all points of P.
    Initialize the triangulation T to contain only T_large.
    for each point p in P do
        Set T ← UpdateDelaunay(T, p).
    end for
    Initialize an empty list T'.
    for each tetrahedron τ in T do
        if τ contains no vertex of T_large then
            Add τ to T'.
        end if
    end for
    return T'.
end procedure
```

Algorithm 4 Computing the Voronoi Diagram from a Delaunay Triangulation

Input: A Delaunay triangulation T of a point set P in $\mathbb{R}^3$.

Output: The Voronoi diagram V of the point set P.

```
procedure DelaunayToVoronoi(T)
    Initialize an empty Voronoi diagram V.
    Initialize maps from tetrahedra to Voronoi vertices, from Delaunay edges to incident tetrahedra, and from Delaunay vertices to incident tetrahedra.
    for each tetrahedron τ in T do
        Compute the circumcenter of τ and add it as a vertex in V.
        Record the circumcenter associated with τ.
        For each vertex of τ, record that τ is incident to that vertex.
        For each edge of τ, record that τ is incident to that edge.
    end for
    for each tetrahedron τ in T do
        for each face κ of τ do
            Let t′ be the tetrahedron adjacent to τ across κ.
            If t′ exists, add an edge in V between the circumcenters of τ and t′.
        end for
    end for
    for each edge α in T do
        Let t_1, t_2, ..., t_k be the tetrahedra incident to α.
        Order t_1, t_2, ..., t_k by turning around α through adjacent faces.
        Let v_1, v_2, ..., v_k be the circumcenters of t_1, t_2, ..., t_k.
        Add the finite chain v_1, v_2, ..., v_k as the Voronoi face dual to α.
        Add edges (v_i, v_{i+1}) for each consecutive pair.
        If the first and last tetrahedra are also adjacent around α, add the closing edge (v_k, v_1).
    end for
    for each vertex p in T do
        Let t_1, t_2, ..., t_m be the tetrahedra incident to p.
        Identify all Delaunay edges incident to p.
        Collect the Voronoi faces dual to those edges.
        Add a cell in V using the vertices contained in those faces.
    end for
    return the Voronoi diagram V.
end procedure
```

Algorithm 5 Setting Up a Reciprocal Lattice

Input: A Bravais lattice type L, lattice parameters $a, b, c, \alpha, \beta, \gamma$, and an index range r.

Output: A finite reciprocal lattice point set P, together with direct basis vectors a_1, a_2, a_3 and reciprocal basis vectors b_1, b_2, b_3.

```
procedure SetupReciprocalLattice(L, a, b, c, α, β, γ, r)
    Normalize the lattice type L to one of the 14 Bravais lattice types.
    Construct primitive direct basis vectors a_1, a_2, a_3 from the lattice type and parameters.
    Compute the cell volume V = a_1 · (a_2 × a_3).
    Compute reciprocal basis vectors
    b_1 = 2π (a_2 × a_3)/V, b_2 = 2π (a_3 × a_1)/V, b_3 = 2π (a_1 × a_2)/V.
    Initialize an empty point set P.
    for i = −r to r do
        for j = −r to r do
            for k = −r to r do
                Let p = i b_1 + j b_2 + k b_3.
                Add p to P.
            end for
        end for
    end for
    Apply a small deterministic perturbation to the points of P.
    return P, (a_1, a_2, a_3), (b_1, b_2, b_3).
end procedure
```